Medical Image Classification Using Deep Convolution Neural Networks

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***Abstract*—This report presents a custom CNN for classifying chest X-rays into Normal and Pneumonia. With preprocessing, augmentation, and hyperparameter tuning, the model achieved 91% accuracy and 0.96 AUC, with reduced false negatives. Compared with ResNet50 and VGG16, the custom CNN delivered the best overall performance for this dataset.**

Keywords—Pneumonia Detection, Chest X-ray Classification, Deep Learning, Convolutional Neural Network (CNN), Hyperparameter Tuning

I. Introduction

Medical imaging plays a vital role in computer-aided diagnosis systems. Pneumonia detection using chest X-ray images can be automated using Deep Convolutional Neural Networks (CNNs). In this work, a custom CNN model is designed to classify chest X-ray images into Normal and Pneumonia classes.

A comparative study is also performed using a pre-trained ResNet50 model. Model performance is evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and AUC-ROC. Hyperparameter tuning is conducted to study model generalisation and overfitting behaviour.

II. Dataset

The dataset used in this study is the Chest X-Ray Pneumonia dataset available on Kaggle.

Dataset Link:

https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

The dataset contains labelled chest X-ray images categorised into:

- Normal
- Pneumonia

A. Dataset Structure

The dataset is divided into three main folders:

- Training set
- Validation set
- Test set

Each folder contains subdirectories corresponding to the two classes.

B. Image Characteristics

- Images are grayscale chest X-rays.
- Resized to 224 × 224 pixels for model input.
- Converted to 3-channel format.

C. Data Preprocessing

The following preprocessing techniques were applied:

1. Rescaling
   All images were normalised by rescaling pixel values from [0, 255] to [0, 1].

2. Data Augmentation (Training Set Only)
   To improve generalisation and reduce overfitting, the following augmentation techniques were applied:
   - Rotation (±15 degrees)
   - Zoom (up to 20%)
   - Width and height shifts (10%)
   - Horizontal flipping
   - Nearest fill mode

Validation and test sets were only rescaled without augmentation.

III. Hyperparameters

Hyperparameter tuning plays a critical role in improving model performance and preventing overfitting or underfitting.

The final selected hyperparameters after experimentation are:

- Image Size: 224 × 224
- Batch Size: 16
- Epochs: 25
- Learning Rate: 0.0001
- Dropout Rate: 0.4
- L2 Regularisation (λ): 0.0007
- Patience: 8

IV. Convolution Neural Network Architecture

A custom Convolutional Neural Network (CNN) was designed for binary classification.

Architecture Overview

The network consists of:

- Three Convolutional Blocks

- Fully Connected Layer
- Dropout Layer
- Output Layer (Sigmoid activation)

A. *Detailed Architecture*

1. Input Layer
    a. Shape: (224, 224, 3)
2. Convolutional Block 1
    a. Conv2D (32 filters, 3×3, ReLU)
    b. L2 Regularization
    c. MaxPooling (2×2)
3. Convolutional Block 2
    a. Conv2D (64 filters, 3×3, ReLU)
    b. L2 Regularization
    c. MaxPooling (2×2)
4. Convolutional Block 3
    a. Conv2D (128 filters, 3×3, ReLU)
    b. L2 Regularization
    c. MaxPooling (2×2)
5. Flatten Layer
    a. Converts feature maps to 1D vector
6. Fully Connected Layer
    a. Dense (128 neurons, ReLU)
    b. L2 Regularization
7. Dropout Layer
    a. Dropout rate = 0.4
8. Output Layer
    a. Dense (1 neuron)
    b. Sigmoid activation (binary classification)

B. *Design Justification*

1) Increasing filter sizes allows hierarchical feature learning.
2) MaxPooling reduces spatial dimensions and computational cost.
3) L2 regularization stabilizes training.
4) Dropout improves generalisation.
5) Sigmoid activation is appropriate for binary classification.

V. Experimental Analysis

A. *Training Set 01*

The hyperparameters used in this training set were:

1. Image Size: 224 × 224
2. Batch Size: 32
3. Epochs: 20
4. Learning Rate: 0.0001
5. Dropout Rate: 0.5
6. L2 Regularisation (λ): 0.001
7. Patience: 5

Model Performance Evaluated

1. Test Accuracy: 86%
2. Precision: 94%
3. Recall: 81%
4. F1 Score: 87%
5. AUC: 94%
6. Confusion Matrix:
    a. True Positive: 319
    b. True Negative: 215
    c. False Positive: 19
    d. False Negative: 71

The recall (0.82) and the relatively high number of false negatives (71 cases) suggest that some true abnormal X-rays are being missed, which could be clinically significant depending on the use case. The gap between training and validation accuracy, along with fluctuating validation loss, indicates moderate overfitting and unstable generalisation. To resolve this, the batch size was reduced from 32 to 16, which can help improve generalisation by introducing more variability during training, leading to more stable validation performance and potentially better recall.

B. *Training Set 02*

The hyperparameters used in this training set were:

1. Image Size: 224 × 224
2. Batch Size: 16
3. Epochs: 20
4. Learning Rate: 0.0001
5. Dropout Rate: 0.5
6. L2 Regularisation (λ): 0.001
7. Patience: 5

Model Performance Evaluated

1. Test Accuracy: 91%
2. Precision: 92%
3. Recall: 92%
4. F1 Score: 92%
5. AUC: 96%
6. Confusion Matrix:
    a. True Positive: 361
    b. True Negative: 206
    c. False Positive: 28
    d. False Negative: 29

After reducing the batch size to 16, the X-ray classification model shows clear improvement, achieving 91% accuracy with strong and well-balanced performance (precision 0.93, recall 0.93, F1-score 0.93). Both classes perform consistently, and the confusion matrix indicates a significant reduction in false negatives (29), which is especially important in medical imaging to avoid missed abnormal cases. Although validation accuracy still fluctuates across epochs, overall generalisation has improved compared to the previous configuration. Further tuning of the dropout rate may help optimise the level of regularisation and enable the model to learn more robust feature representations while maintaining generalisation.

C. *Training Set 03*

The hyperparameters used in this training set were:

1. Image Size: 224 × 224
2. Batch Size: 16
3. Epochs: 20
4. Learning Rate: 0.0001
5. Dropout Rate: 0.4
6. L2 Regularisation (λ): 0.001
7. Patience: 5

Model Performance Evaluated

1. Test Accuracy: 89%
2. Precision: 93%
3. Recall: 87%

4. F1 Score: 90%
5. AUC: 95%
6. Confusion Matrix:
    a. True Positive: 341
    b. True Negative: 212
    c. False Positive: 22
    d. False Negative: 49

Reducing the dropout rate to 0.4 led to a slight decline in performance, with accuracy dropping to 89% and false negatives increasing to 49, indicating more missed abnormal X-rays. Although precision (0.94) and AUC (0.95) remain strong, recall decreased, and validation performance was unstable. Further, the L2 regularisation parameter was adjusted; the goal was to fine-tune the strength of regularisation to improve generalisation while maintaining model capacity.

D. Training Set 04

The hyperparameters used in this training set were:

1. Image Size: 224 × 224
2. Batch Size: 16
3. Epochs: 20
4. Learning Rate: 0.0001
5. Dropout Rate: 0.4
6. L2 Regularisation (λ): 0.0007
7. Patience: 5

Model Performance Evaluated

1. Test Accuracy: 89%
2. Precision: 92%
3. Recall: 90%
4. F1 Score: 91%
5. AUC: 95%
6. Confusion Matrix:
    a. True Positive: 353
    b. True Negative: 205
    c. False Positive: 29
    d. False Negative: 37

After reducing L2_LAMBDA to 0.0007, the model maintained stable performance with 89% accuracy, strong precision (0.92), improved recall (0.91), and a high AUC (0.95), while reducing false negatives to 37, which is important for minimising missed abnormal X-rays. Further, the Patience and number of Epochs were increased to give the model more time to converge and potentially capture additional validation improvements, while monitoring for signs of overfitting.

E. Training Set 05

The hyperparameters used in this training set were:

1. Image Size: 224 × 224
2. Batch Size: 16
3. Epochs: 25
4. Learning Rate: 0.0001
5. Dropout Rate: 0.4
6. L2 Regularisation (λ): 0.0007
7. Patience: 8

Model Performance Evaluated

1. Test Accuracy: 91%
2. Precision: 93%
3. Recall: 91%
4. F1 Score: 92%
5. AUC: 96%
6. Confusion Matrix:
    a. True Positive: 355
    b. True Negative: 210
    c. False Positive: 24
    d. False Negative: 35

With patience increased to 8 and epochs to 25, the model shows improved overall performance. Accuracy increased to 91%, precision remains high at 0.94, recall improved to 0.91, and the AUC reached 0.96, indicating strong discriminative ability. False negatives decreased slightly to 35, which is important for reducing missed abnormal X-rays in a clinical setting. Although validation accuracy still fluctuates across epochs, the final test performance is stronger and more balanced compared to earlier configurations.

VI. Comparison with Pre-trained Models

To evaluate the effectiveness of the proposed custom CNN architecture, its performance was compared against two widely adopted pre-trained deep convolutional models: ResNet50 and VGG16, both fine-tuned on the same dataset. The comparison highlights differences.

TABLE I. Comparison with Pre-trained Models

Model	Accuracy	Precision	Recall	F1-Score	AUC
ResNet50	0.76	0.7565	0.9000	0.8220	0.8534
VGG16	0.88	0.8962	0.9077	0.9019	0.9433
Custom CNN	0.91	0.9300	0.9100	0.9200	0.9600

A. ResNet50

Confusion Matrix

- True Positive: 351
- True Negative: 121
- False Positive: 113
- False Negative: 39

Performance Summary

ResNet50 shows high recall but significantly lower precision and accuracy, mainly due to a high number of false positives.

The model tends to misclassify Normal X-rays as Pneumonia, indicating difficulty in distinguishing subtle normal features. This behaviour is expected from deeper architectures like ResNet50, which typically require larger datasets to generalise well.

As a result, ResNet50 demonstrates weaker overall performance and stability compared to the custom CNN.

B. VGG16

Confusion Matrix

- True Positive: 354
- True Negative: 193
- False Positive: 41
- False Negative: 36

Performance Summary

VGG16 comes close to the Custom CNN, but the Custom CNN still leads in all evaluation metrics.

Compared with VGG16, the Custom CNN has higher accuracy, higher F1-score, and the best AUC value, meaning it separates the two classes more effectively.

The confusion matrix also shows that the Custom CNN has fewer false positives and false negatives than VGG16.

VGG16 learns the X-ray patterns well, but the Custom CNN is more finely tuned for this specific task, giving it a slight advantage.

VII. Sample Dataset Images

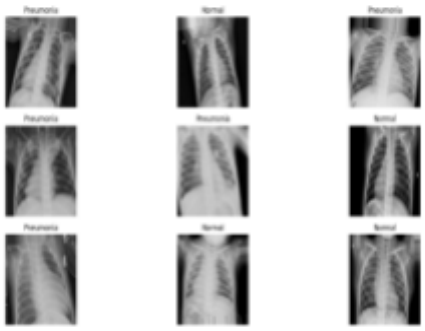

Fig.1. Sample Dataset Images

VIII. Prediction Results

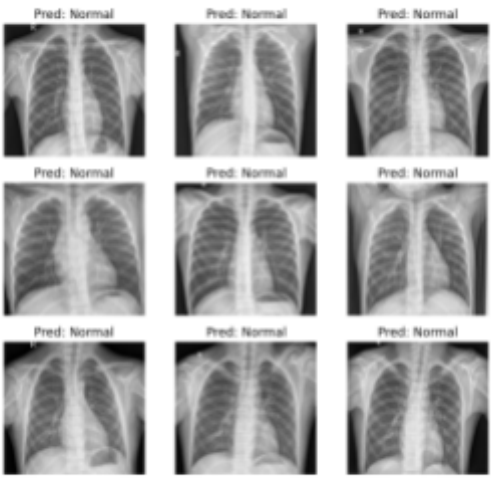

Fig.2. Prediction Results

IX. Training vs Validation Loss and Accuracy Curves

A. Training vs. Validation Loss

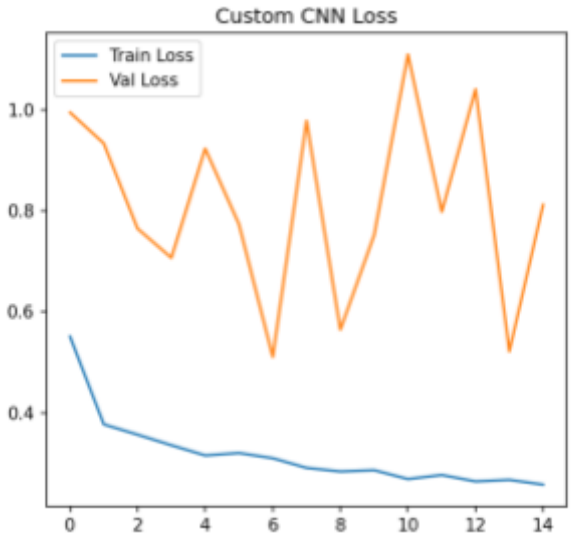

Fig.3. Training Vs Validation Loss

Training loss consistently decreases, which shows the model is fitting the training data effectively over epochs.

Validation loss remains high and highly unstable with frequent spikes, indicating overfitting and weak generalisation on unseen data.

B. Training vs. Validation Accuracy

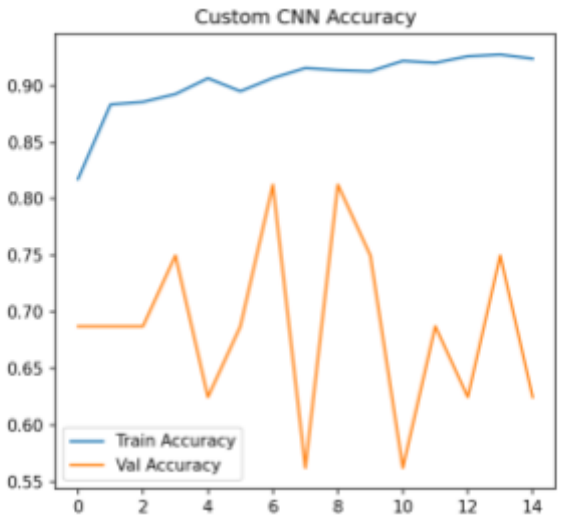

Fig.4. Custom CNN Accuracy

Training accuracy steadily improves and stabilises above 0.90, reflecting strong learning on the training set.

Validation accuracy fluctuates widely (roughly 0.56 to 0.81), suggesting inconsistent validation performance and the need for stronger regularisation or data balancing.